TriOPS: Web-Based QR Production Process with Salary Management System and Workforce Analytics in Tagum Trio Lumber Corporation

A Capstone Project Presented to
Faculty of the Institute of Computing
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Chapter I
INTRODUCTION

A. Background of the Study

In plywood manufacturing, the transformation of raw materials into finished products involves multiple interconnected processes such as log receiving, veneer cutting, drying, pressing, and finishing. However, many manufacturing companies still rely on manual monitoring systems, resulting in inefficiencies, delays, and limited real-time visibility of operations [1]. The transition toward digital and automated systems is essential for improving manufacturing performance under Industry 4.0. It emphasized the connected and smart systems enable real-time data collection, which enhances operational efficiency and decision-making in industrial environments [2].

The Department of Science and Technology (DOST), through the Harmonized National Research and Development Agenda (HNRDA), the role of science, technology, and innovation in enhancing industrial productivity and ensuring that research outputs generate economic and social benefits for the country [3], Real-time production monitoring systems have been widely recognized for their ability to improve manufacturing performance. Studies show that such systems help identify bottlenecks, reduce downtime, and enhance productivity by providing immediate feedback on production activities [4]. Furthermore, integrating labor monitoring with production systems improves workforce management and accountability. Performance-based compensation systems can increase worker productivity when supported by accurate tracking systems. However, manual payroll computation often leads to errors and delays, highlighting the need for automated payroll systems integrated with production data [5].

At the institutional level, the proposed study aligns with the Davao del Norte State College (DNSC) Research, Development, and Extension (RDE) Agenda, which focuses on sustainable development, innovation, and economic productivity within the region. The DNSC framework emphasizes that research initiatives should contribute to environmental conservation, community resilience, and economic productivity, in line with the Sustainable Development Goals [6].

This study ensures alignment with value chain optimization. Enterprise 4.0 refers to a new generation of connected and intelligent factories. It is a digital transformation characterized mainly by intelligent automation and the integration of new technologies into the enterprise value chain, giving life to an interconnected enterprise 4.0 where machines, products, and people interact [7].

This is supported by local studies in the Philippines which highlight that digital transformation and information systems significantly improve operational efficiency and decision-making in small and medium enterprises (SMEs). These findings reinforce the relevance of implementing real-time monitoring and payroll systems in manufacturing settings such as plywood production.Manufacturing industries, particularly those involved in wood processing and plywood production, play a significant role in economic growth and industrial development. These industries align with the United Nations Sustainable Development Goals (SDGs), particularly SDG 9 (Industry, Innovation, and Infrastructure), which promotes sustainable industrialization and the adoption of innovative technologies to improve efficiency and productivity [8].

B. Company Overview

Tagum Trio Lumber Corporation is a prominent plywood manufacturing enterprise established in June 2025 located at Brgy Pinagkaisa Lapaz Carmen Davao del Norte and is currently under the capable management of Mr. Jiang Kuanrong. As the company's head, Mr. Jiang oversees the overall operations and spearheads the strategic direction of the business, ensuring sustainable growth and operational excellence. As a developing manufacturing firm, the company is primarily engaged in the production of high-grade plywood products designed for various applications, including residential and commercial construction, furniture making, cabinetry, and other industrial uses. With a substantial workforce of approximately 300 employees, the company plays a vital role in the local economy by providing significant employment opportunities and actively supporting economic activity within the region.

As shown in the figure 1 below, TriOPS operates under a well-defined hierarchical structure where the Chief Executive Officer sets the overall direction of the company and delegates operational authority to the General Manager, who bridges leadership and all departments to ensure organizational goals are met. The Accounting department maintains the company's financial health by managing transactions, finances, and timely payroll, contributing to workforce stability and morale. The HR department, though compact, plays a vital role in personnel management, workplace compliance, and maintaining a productive work environment. The Production department, being the most operationally intensive, follows a clear chain of supervision from the Production Supervisor to the Production In-Charge and down to the Production Leadmen, who serve as the backbone of daily operations ensuring consistent and quality output. Overall, each role is purposefully structured to complement one another, allowing every department to function independently yet collaboratively, driving the organization toward sustained efficiency and long-term success.

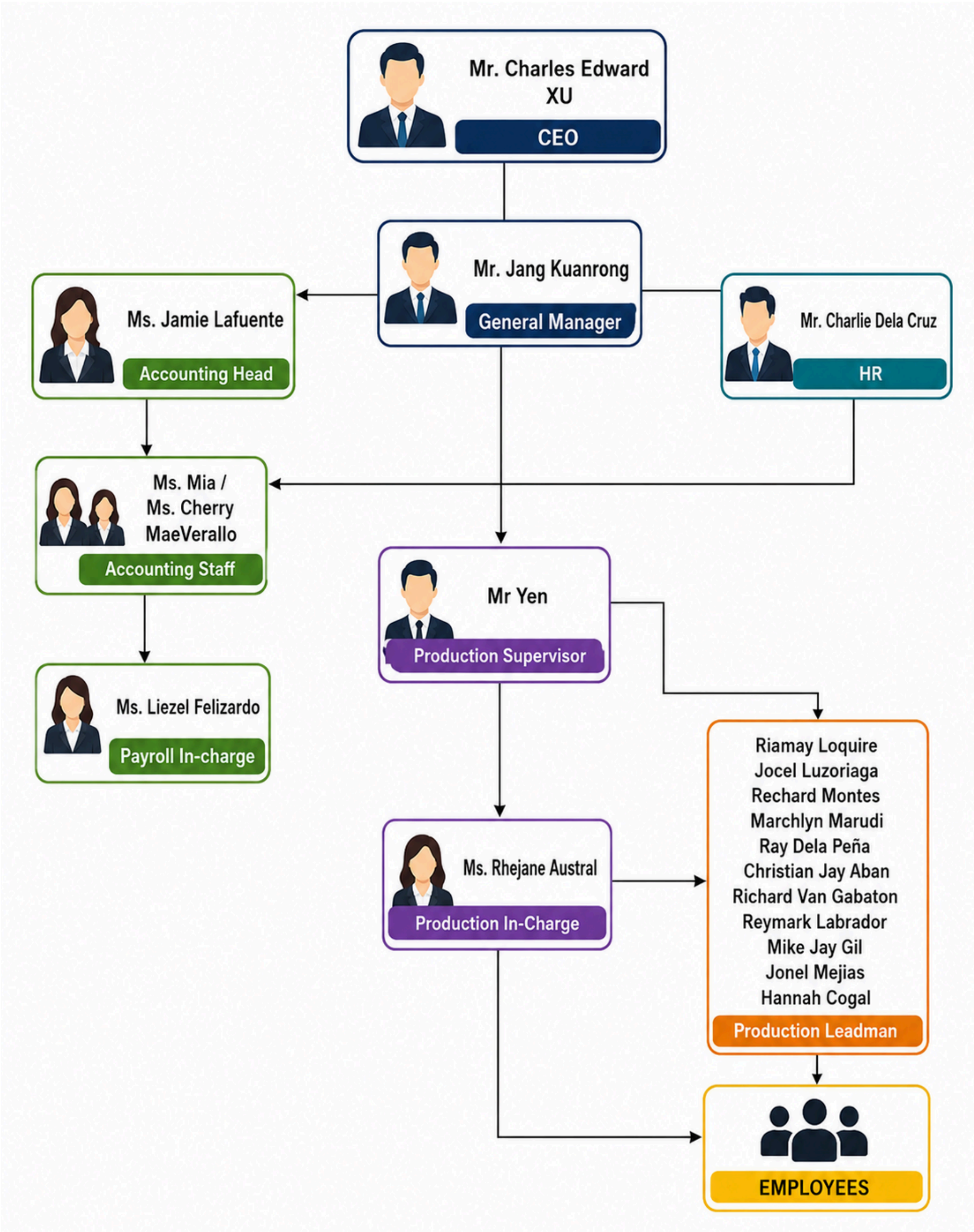

Figure 1. Organizational Structure

C. Problems Encountered

The Production Head of Tagum Trio Lumber Corporation is responsible for overseeing approximately 300 employees within the production department alone. Currently, the manual management of essential tasks such as maintaining employee records, processing salaries, tracking leave requests, handling excused absences, and monitoring half-day schedules has become a significant challenge. This outdated method places a heavy administrative burden on the team and creates unnecessary complexities. Consequently, it leads to delays, reduces overall efficiency, and significantly slows down daily operational processes. The key problems encountered by Tagum Trio Lumber Corporation are as follows:

- Inefficient data management due to reliance on paper records and manual entry.
- Time-consuming preparation of production and payroll reports.
- Lack of material traceability from raw state to finished goods.
- Delayed decision making due to lack of real-time production data.
- Unclear performance evaluation with no standard way to compare production rates and workloads.
- Difficulties in tracking staff deployment and transfers between departments.

D. Existing Business Process

This section presents the existing business processes of Tagum Trio Lumber Corporation, with specific focus on the core human resource functions that serve as the backbone of the company's daily operations and long-term management. These key functions include employee records management, attendance monitoring, payroll processing, and employee performance evaluation. These processes were identified and selected for analysis because they represent the most critical, high-volume, and complex activities within the HR department—activities that directly influence workforce productivity, operational efficiency, and the quality of management decisions across the organization.

At present, these processes are primarily executed through manual and semi-automated methods. Operations rely heavily on physical documentation, logbooks, Excel spreadsheets, and raw or unprocessed data outputs from biometric devices. Since these systems operate independently and are not centralized or integrated, the management of these workflows has become increasingly challenging. These methods are not only time-consuming and labor-intensive but also highly susceptible to human error, data inconsistencies, and information loss. This limitation becomes even more pronounced as the company continues to expand its workforce and maintain operations across different work sites, departments, or facilities. The following subsections illustrate each process through process flow diagrams and provide comprehensive, step-by-step explanations of how these workflows are currently performed.

Record Product Process

This section details the production process at Tagum Trio Lumber Corporation, focusing on the departmental structure, workforce organization, and the critical role of leadmen in overseeing operations and managing labor. The Production Department is segmented into multiple specialized units, each designed to handle distinct stages or aspects of the lumber production cycle. Each of these production units is staffed by a team of 10 to 15 skilled workers, each assigned specific roles and responsibilities that contribute to the overall efficiency and output of their respective department. This departmentalization allows for focused expertise and optimized task allocation within the complex lumber manufacturing process.

Supervising these specialized units are Leadmen, who play a pivotal role in maintaining operational flow and ensuring accountability. While each departmental unit has its dedicated workforce, the Leadmen are tasked with a broader scope of responsibility, often overseeing multiple departments. This multi-departmental oversight requires Leadmen to meticulously track and record the service delivery, productivity, and various operational metrics across the departments under their charge. Their primary function includes monitoring daily progress, addressing immediate production challenges, ensuring adherence to quality standards, and accurately documenting the work performed by each team. This comprehensive approach to supervision by the

Leadmen is essential for integrating departmental efforts and for providing crucial data for overall production planning and performance evaluation.

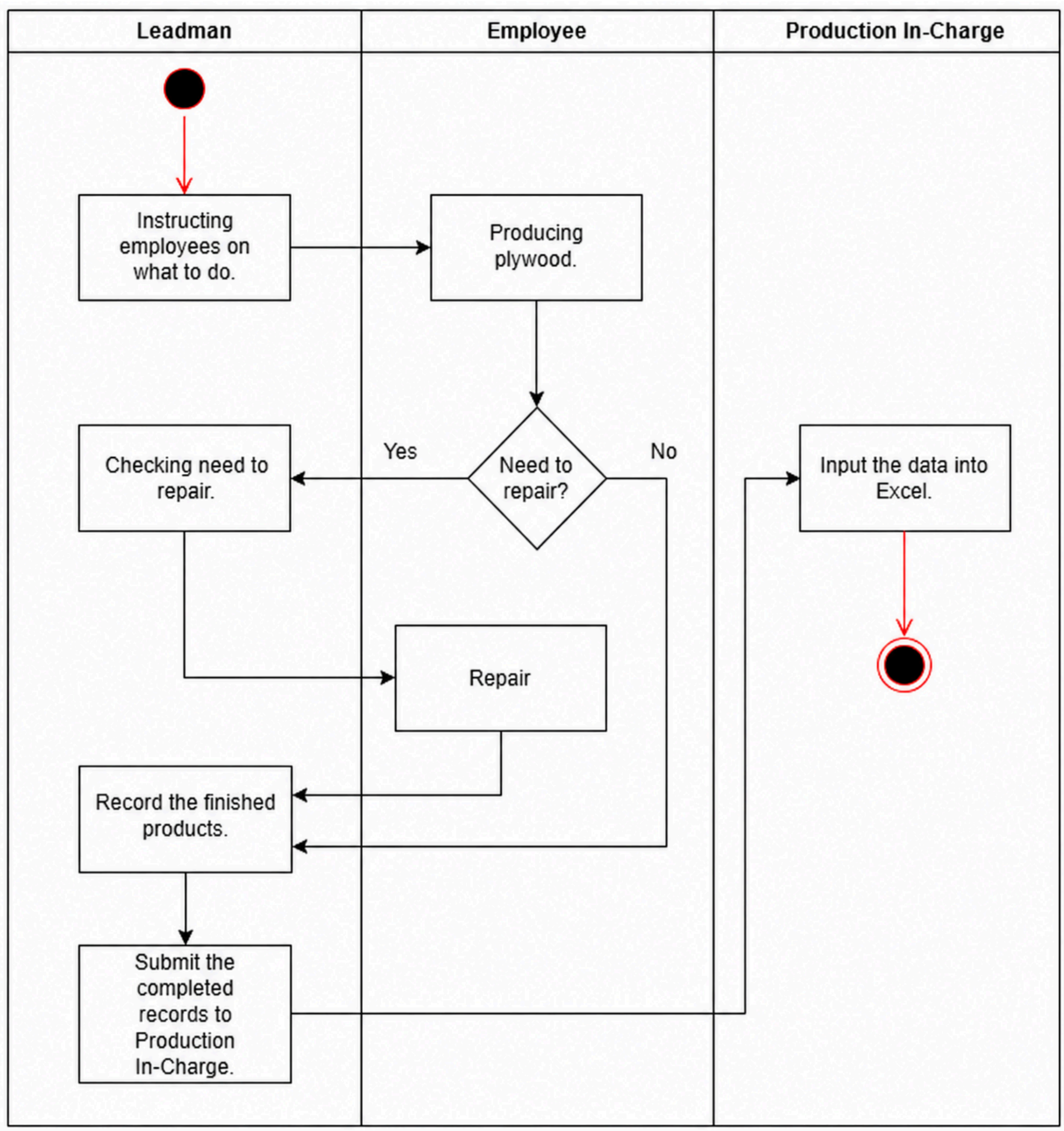

Figure 2. Current Production Process(dili production head it should be Production In-Charge)

Salary Process

The figure below illustrates the current process of distributing and calculating the salaries of Tagum Trio employees. The salary is based on a piece-rate system, where employees are paid according to their output. The production head is responsible for calculating each employee’s salary in the production department using daily production records. The total salary is determined by multiplying the number of items produced by the corresponding rate of each product, which may vary depending on its width, size, and thickness.

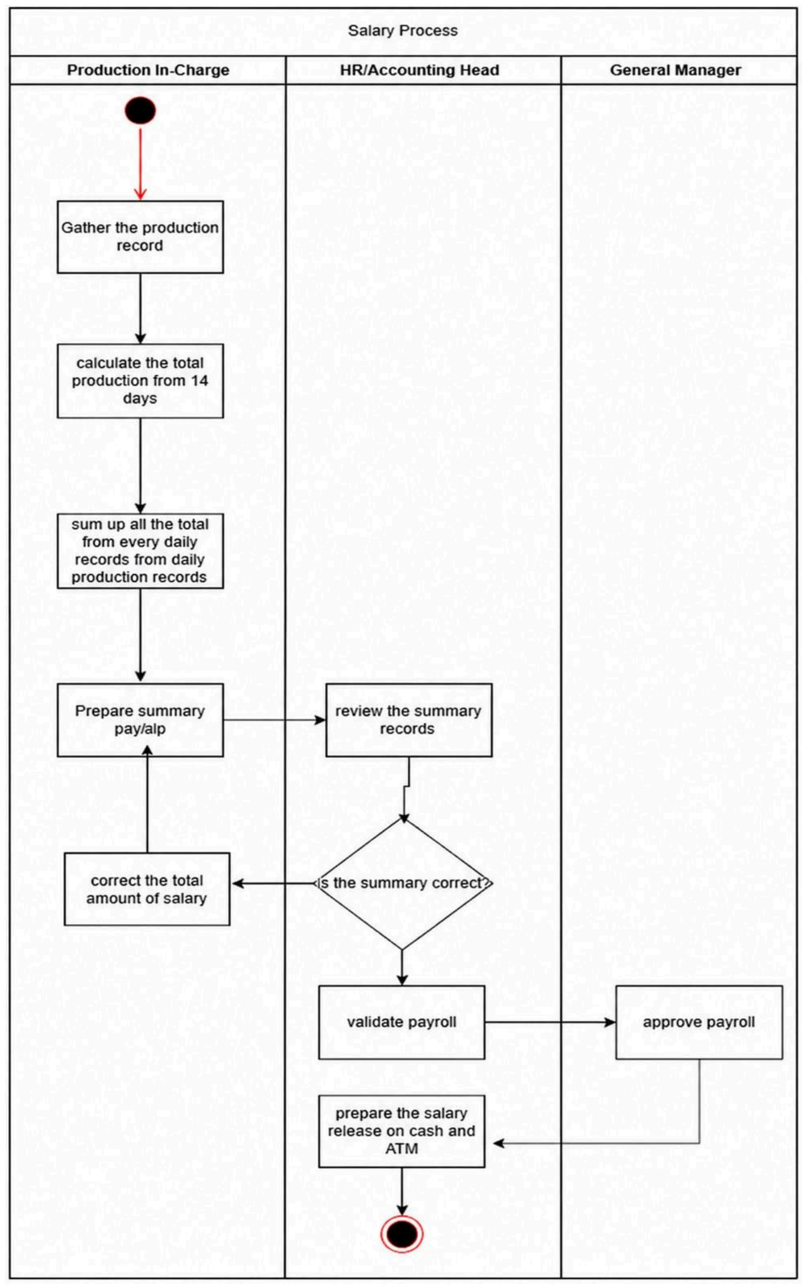

Figure 3. Current Payroll Processing(dili production head it should be Production In-Charge)

Reassign Department Processes

The provided figure illustrates the employee reassign process. , only the CEO and Manager possess the authority to reassign employees between production departments. These Reassigning are typically based on their observations; if a department is understaffed, employees may be reassigned to cover the shortfall. Currently, these Reassigning are conducted verbally, lacking proper documentation. While a leadman can assign tasks to these temporarily reassigned employees, their compensation is calculated based on their individual output within that department. The reassigned employee then returns to their original department and position the following day.

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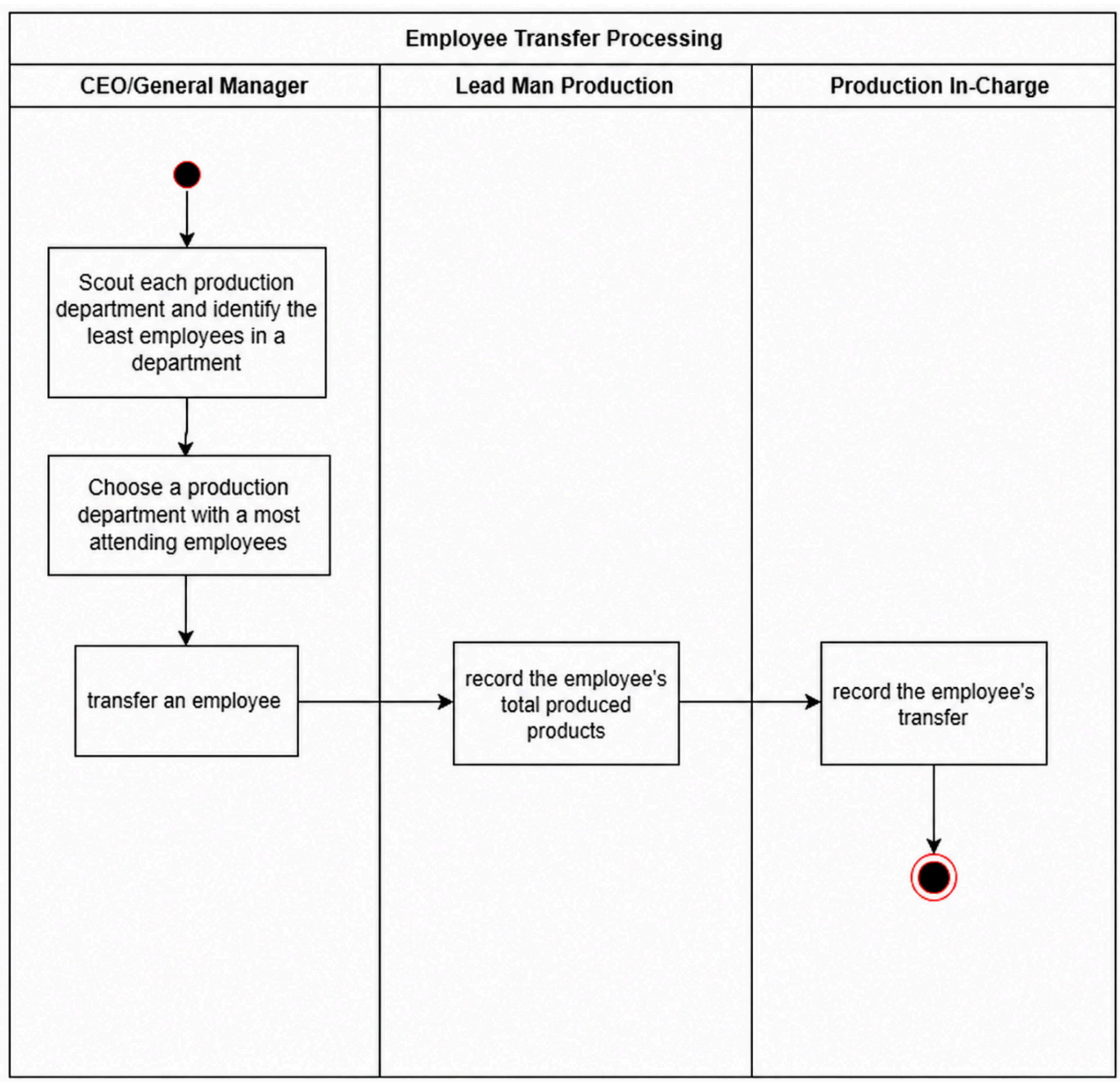

Figure 4. Current Reassign Department Processes

Leave Processes

The illustration below details the company's leave approval process. A key step in this procedure, which typically extends the approval time to 2-3 days, involves the translation of the employee's leave request letter. This translation is essential to ensure that the General Manager or CEO, who may not be proficient in the original language of the request, can fully understand the contents and make an informed decision regarding the leave application.

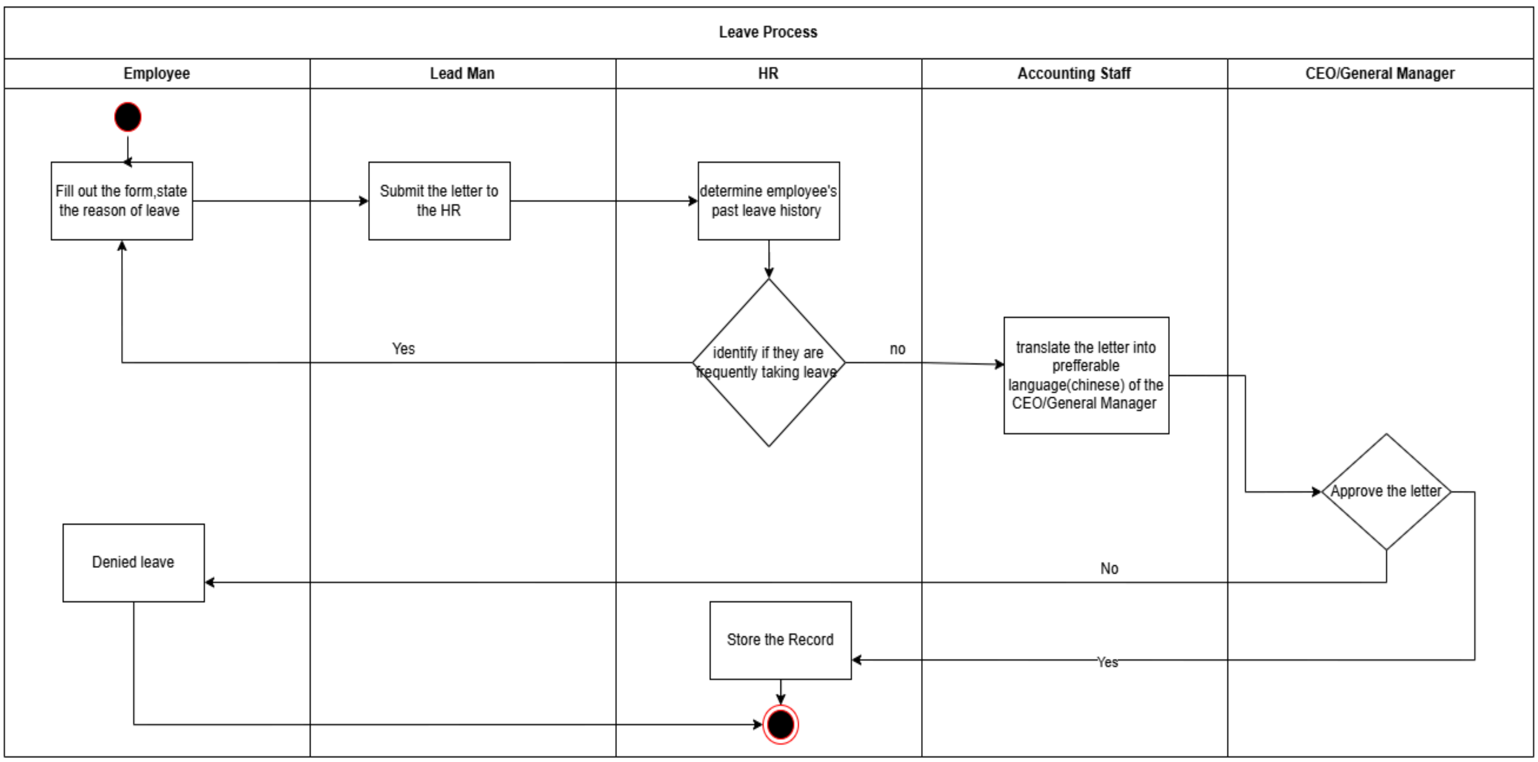

Figure 5. Current Leave Processes

E. Objectives of the Project

a.General Objectives

The overall objective is to develop an integrated website system that centralizes data management, automates production recording and compensation calculations, streamlines operational processes, and facilitates clear communication and access controls, all to improve efficiency, accuracy, and transparency across the organization.

b.Specific Objectives

The following specific objectives aims to:

1. Implement an admin control module for secure login, credential management, user access rights, and system settings.
2. Develop a centralized employee information management module that securely stores, updates, and manages employee records, including personal details, work history, and supporting documents.
3. Establish a production oversight and management module for administrators to monitor real-time progress, identify operational concerns, and manage production sections.
4. Design a digital request management system that enables employees to submit various applications and allows administrators to review and approve employee compensation, work logs, and transfers.
5. Integrate a QR Code-based Production and Transfer Management System that allows leadmen to scan finished products for automatic record updates and calculation of employee salaries based on output, and enables employees to request to be reassign to officially log new transfers from different departments.
6. Implement an employee communication and self-service portal for employees to access reminders, schedules, official announcements, and view their assigned sections and daily work details.
7. Provide a finance module for the finance department to review, approve, and generate reports for payslips.
8. Establish a comprehensive Workforce Analytics Platform that leverages integrated data from attendance, production, and employee transfers to generate insightful reports and dashboards, enabling management to optimize staffing, enhance productivity, identify training needs, and make data-driven decisions on workforce allocation and performance.

F. Scope and Limitation

This study focuses on the design and development of a comprehensive Web-Based QR Production Process with Salary Management System and Workforce Analytics in Tagum Trio Lumber Corporation. This system is specifically engineered to support and enhance crucial operational and human resource processes within the plywood manufacturing enterprise, including secure admin control, centralized employee information management, real-time production oversight, streamlined digital request management, QR code-based production and reassigning tracking, employee communication and self-service, finance module functionality, and a robust workforce analytics platform.

In terms of people**,** the system will be utilized by General Manager, HR personnel, finance department staff, and leadmen production. All employees of Tagum Trio Lumber Corporation will interact with the system through the self-service portal to access information and submit various applications.

In terms of hardware**,** the system will operate on standard computing devices such as desktop computers and laptops, readily available within the corporation. It will seamlessly integrate with QR code scanning devices to facilitate production and transfer management.

In terms of software, the Web-Based QR Production Process with Salary Management System and Workforce Analytics will be developed as a modern web-based application. The frontend will leverage cutting-edge technologies such as Tailwind CSS, and React.Native for UI a responsive and intuitive user experience.

In terms of data**,** the system will securely manage a wide array of information crucial to Tagum Trio Lumber Corporation's operations. This includes comprehensive employee records (personal details, work history, supporting documents), real-time production progress data, digital request logs (leave, overtime, concerns), QR code scan data for finished products and employee transfers, attendance records, and payroll-related data. The integrated Workforce Analytics Platform will process and analyze these datasets to generate insightful reports, dashboards, and visualizations, empowering management with data-driven decision-making capabilities.

In terms of process**,** the Web-Based QR Production Process with Salary Management System and Workforce Analytics within Tagum Trio Lumber Corporation will automate and streamline key operational and HR workflows. This encompasses secure login and user access rights management, efficient employee information management, real-time monitoring of production sections, a paperless digital request system, automated calculation of employee salaries based on scanned output, and seamless tracking of employee transfers viaRequesting to transfer in other department. The finance module will facilitate review, approval, and generation of payslips. Furthermore, the Workforce Analytics Platform will provide comprehensive insights into staffing, productivity, training needs, workforce allocation, and performance optimization.

This study is specifically limited to the design and implementation of the proposed Web-Based QR Production Process with Salary Management System and Workforce Analytics within Tagum Trio Lumber Corporation. It does not encompass integration with external systems such as banking systems for salary disbursement or government systems for statutory compliance. The system relies on the accurate input of data from QR code scans and manual entries; thus, any incorrect or incomplete data may impact the accuracy of system outputs. Additionally, the system does not include advanced hardware-level modifications for QR code scanners. The Workforce Analytics Platform is designed around predefined criteria and rules set by the organization and does not currently utilize advanced artificial intelligence or machine learning algorithms for predictive analysis. The system is intended for internal use only and requires either internet or local network connectivity for access.

G. Related Literature

Qr Production Process

In the old document management system, information was entered manually into the logbook, which wasted time and increased the likelihood of errors in recording information. This process is inefficient and unsuitable for larger companies or institutions with a significant number of documents. Since many employees tend to lose track of document paths, these systems should have the capability to trace the journey of documents from their source to their destination [9].

QR codes, widely used for document tracking, are compact, square-shaped barcodes that can be scanned effortlessly with a smartphone or other QR scanner. This allows users to quickly find information or records related to the code. QR codes offer many benefits for document tracking, can involve expedient and effortless access to, as well as sharing of, information, and they easily integrate with existing records management systems [10]

Salary Management

Every organisation keeps a record of their staff. Staff records play a crucial role in staff management. Every organisation requires these records to calculate pay, manage workforce and see performance of employees [11].

Management of all these records is a challenging task and time consuming process for the HR team [12] , which can be reduced by using EMS, which is the Employee Management System [13].

The implementation of web-based salary management systems can address these challenges by automating processes, enhancing real-time data access, and ensuring better security measures. A growing body of research highlights the significant advantages of adopting such systems, with several studies focusing on the use of advanced frameworks like the Zachman Framework in developing robust, scalable, and integrated solutions for public institutions[14].

Workforce Analytics

Data has become a critical asset in the modern business landscape, driving decision-making processes across industries. Organizations increasingly rely on advanced analytical tools to gain insights that enhance efficiency, foster innovation, and sustain competitiveness [15].Workforce analytics, which focuses on leveraging data to optimize human resource management, provides organizations with the means to understand and address complex workforce challenges. It enables the strategic alignment of talent acquisition, performance management, and employee retention with broader business objectives, fostering a dynamic and adaptable workforce [16].

H. Related Systems

Qr Production Process

The University of the Philippines (UP) employs a document tracking system (DTS) for enhanced efficiency and transparency [17]. The UP DTS lacks route functionality, causing potential delays [18]. Incorporating routes would significantly improve its effectiveness. QR codes, increasingly popular due to their touchless nature, connect physical and digital realms [19]. Lingaya suggests a tracking model for Philippine education institutions, easing document management [20].

Salary Management

Numerous studies have been conducted exploring various aspects of payroll systems and utilizing diverse techniques. Singh et al. [21] presented a study titled "automated payroll system (A-PAY)," aimed to digitize the payroll process for faculty members, including the management of their personnel information and the calculation of allowances and deductions. The study utilized Vb.net as the front end and Microsoft Access 2007 with SQL Server 2008 as the back end. The result was a more efficient system with options for data backup and restoration, improved security, reduced calculation errors, and automatic report generation.

Hikmah and Muqorobin [22] developed an employee payroll information system for a Consulting Engineering Services company. They aimed to eliminate the challenges of manual payroll calculation. They conducted research through observation, interviews, documentation, and literature review, and used PHP and a MySQL database to build the system. The design improved payroll management and provided faster and easier retrieval of employee payroll information, leading to improved administrative processing efficiency.

Workforce Analytics

According to Huselid (2018), workforce analytics shifts human resource management from intuition-based decision-making toward evidence-based practices. Its international relevance is grounded in global industrial challenges such as labor shortages, high turnover, and fluctuating productivity levels in manufacturing sectors across both developed and emerging economies. The systematic deployment of workforce analytics is facilitated by Management

Information Systems (MIS), which serve as digital platforms to collect, store, and interpret human capital data [23]

In manufacturing environments, labor planning is a key function of workforce analytics that aims to match human resource capacity with production schedules while minimizing costs and disruptions [24]